

ENVS 423L/EPS 522 Problem Set #1

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by 5pm on Tuesday, August 26 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of some basic hydrologic concepts.

1.1 Unit Conversion

Rob holds a water right for 200 acre-feet per year for his brand new well. What pumping rate can he sustain in gallons per minute and cubic meters per second? If he owns an olympic-sized swimming pool, how many times could he empty and completely fill it in one year? (20 pts)

1.2 Dimensional Analysis, part 1

Viscosity (μ) describes a fluid's resistance to flow. Given equation 1, where F is force, A is area, u is velocity, and h is height, what are the dimensions of μ ? (20 pts)

$$\frac{F}{A} = \mu \frac{u}{h} \quad (1)$$

1.3 Dimensional Analysis, part 2

The Reynolds number (Re) is a quantity that describes whether fluid flow is laminar or turbulent. Given Equation 2, where ρ is density and L is length, what are the dimensions of Re? (20 pts)

$$Re = \frac{\rho u L}{\mu} \quad (2)$$

1.4 Capillary Rise

Calculate the expected capillary rise for a tube with radius of 0.5 mm and for water at standard temperature and pressure (STP). (15 pts) For the same

radius, what is the expected rise for water at 30 degrees C? Recall both surface tension and density are temperature-dependent. (15 pts)

1.5 Where does your water come from?

Pick your hometown (or any other place you have lived) and describe where your water comes from. If there are multiple sources of water, list them all. (10 pts)